Unit 7: The User Interface

Unit Learning Objectives

1. Explain the concepts of user experience, user interface, and usability
2. Describe the metaphors that can be used to assist in user-interface design
3. Describe important characteristics of human-interface objects that affect usability
4. Be able to design a suitable user interface to successfully complete a use case

Key Concepts and Definitions

Key Terms	Explanation
User Experience (UX)	Applies to all interactions a person might have with a software application Includes actions, responses, perceptions and feelings
User interface (UI)	The set of inputs and outputs a user interacts with Part of the system that the users directly interact with Provides a dialog between actor and system
User-Centered Design	Experienced developers recognize the importance of all aspects of the user interface and recommend using several techniques that place the user at the center of the development process. These techniques are referred to collectively as user-centered design.
Usability	The goal of user-centered design is **usability**, the degree to which a system is easy to use and easy to learn Ease of learning and ease of use sometimes conflict. For example, a system can be easy to learn but be cumbersome to use all day. Usability is not always easy to achieve.
Human-Computer Interaction (HCI)	Field of study that is concerned with the efficiency and effectiveness of user-interfaces

The user experience is a broad concept that applies to all aspects of a person's interaction with the system. The user interface, which is the set of inputs and outputs that the user sees and interacts with, must fit within this larger concept of the user experience.

User interface design must consider all aspects of the user characteristics as well as the numerous devices that the user will interact with. Experienced developers recognize the importance of all aspects of the user interface and recommend using several techniques that place the user at the center of the development process.

These techniques are collectively referred to as ***user-centered design***. User-centered design techniques emphasize three principles to consider during the development of an information system:

- **Focus early on users and their work**
  The early focus on users and their work is consistent with the approach to systems analysis in this text. User-oriented analysis and design tasks are performed as early as possible and are often given higher priority than other tasks.
- **Evaluate designs to ensure usability**
  The goal of user-centered design is usability, the degree to which a system is easy to learn and use. Ease of learning and ease of use sometimes conflict. For example, a system can be easy to learn but be cumbersome to use all day. Usability is not always easy to achieve.
- **Use iterative development**
  The third principle of user-centered design is iterative development—that is, doing some analysis, then some design, then some implementation, and then repeating the processes. After each iteration, the project team evaluates the work on the system to date. Iterative development keeps the focus on the user by continually returning to the user requirements during each iteration and by evaluating the system after each iteration.

User interface design is, in fact, included in a much larger field of **Human Computer Interaction**. This larger field focuses on all aspects of efficiency and effectiveness as humans utilize technology to perform their business tasks

Metaphors of HCI

There are various metaphors of how a user accesses and interacts with an automated system. Metaphors assists analysts to understand, define and describe the user interface more effectively.

Metaphor	Description	Example
Direct manipulation	Manipulating objects on a display that look like physical objects (pictures) or that represent them (icons)	The user drags a folder icon to an image of a recycle bin or trash can to delete a collection of files.
Desktop	Organizing visual display into distinct regions, with a large empty workspace in the middle and a collection of tool icons around the perimeter	At computer startup, a Windows user sees a desktop, with icons for a clock, calendar, notepad, inbox and sticky notes (the computer interface version of a physical Post-It note).
Document	Visually representing the data in files as paper pages or forms; these pages can be linked together by references (hyperlinks)	The user fills in a form field for a product he or she owns, and the manufacturer's Web site finds and displays the product's manual as an Adobe Acrobat file, which contains a hyperlinked table of contents and embedded links to related documents.
Dialog	The user and computer accomplishing a task by engaging in a conversation or dialog by using text, voice, or tools, such as labeled buttons	The user clicks a button labeled "troubleshoot" because the printer isn't working. The computer prints questions on the display, and the user responds by typing answers or selecting responses from a printed list.

Fundamental Principles of User-Centered Design

It is important in the detailed design of the user interface to consider good design principles. Such design principles as visibility, affordance, user shortcuts, feedback, closure, error messages, and reversal of actions are all important no matter what type of user interface is being designed.

The sensible use of menus and navigation techniques also make the user interface more usable. Often the best way to organize menu items is to group menu items together by actor or user role.

Human- Interface Objects (HIO)	Icons and other objects on a screen that can be manipulated by the user and cause some action to occur
Consistency	Refers to the similarity of presentation in different areas of the application
Discoverability	It should be easy for the user to find and figure out the operations and features on a screen Sometimes we say, the application is "intuitive" – meaning is it easy to figure out what to do
Closure	Closure lets the user know that he/she has completed an operation
Readability and Navigation	Make the text readable by type, size, and color Navigation through the entire application should be obvious and easily traversed
Usability and Efficiency	Which simply mean that designers should always be thinking about how to make the application easy to use

Human-Interface Objects (HIO)

Human interface objects are those objects that appear on a screen and that the user can manipulate or use to navigate. When designing the user interface, use an HIO that reflects its purpose or behaviour. Here are some examples of HIOs that suggest its function:



The Save button suggest, when clicked, it will Save work. This is called **Affordance** – the appearance of a specific control, suggest its function

HIO's should also provide visual feedback when they are activated. This feedback can be visual, such as a change in its color or background color or highlight, or audio, such as a clicking noise. When a control is visible so that users know it is available, they have **Visibility.**

Consistency

One of the most difficult approaches during design is a lack of consistency. When different icons or HIOs are used to do the same operation, or when the same HIOs have unique operations, then users really get confused. There are several levels of consistency that are important.

- **Consistency within a platform** is achieved when all the same HIOs are used within the Windows platform or the Apple platform. For general purpose HIOs this is normal because those items, such as buttons, and arrows, etc. are provided by the platform developer. Consistency across platforms, i.e. Windows and Apple, is harder. Some HIOs have consistent use, but not all do.

- **Consistency within a suite of applications** means that the set of related programs all have the same look-and-feel, and use the same HIOs. A common example is Microsoft Office Suite. Most general purpose suites have gone this way, but designers should be sure they do the same.

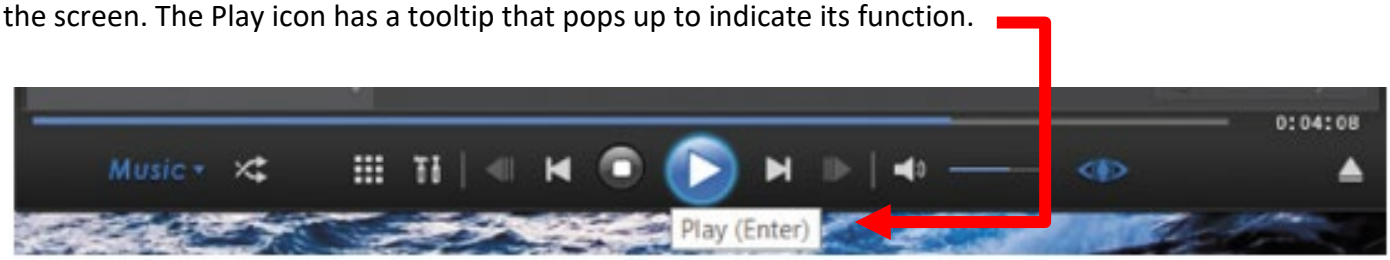
- **Consistency within an application** means that the various screens all use the same HIOs to do the same functions. This principle is frequently violated.

Discoverability

Discoverability is the principle which means that it should be easy for the user to find and figure out the operations and features on a screen.

Active Discovery is simply the principle that the designer will include these kinds of clues by design and not simply let the user find the available functions by trial and error.

Tool tips can help with active discovery by displaying a pop up when a user hovers over icons or hotspots on the screen. The Play icon has a tooltip that pops up to indicate its function.



Other clues might include changing of the mouse pointer, or hover actions such as changing color. Or have little pop-up images or textboxes that help the user discover the allowable functions.

Closure

Closure lets the user know that he/she has completed an operation. Visibility and feedback are important aspects of closure. When using a dialogue metaphor it is especially important to provide closure so that the user knows the job is complete. Dialogues often require several steps, which all must be completed.

Another part of closure is that the user's work should be saved or protected both during a dialogue, but especially at the end. That can be done either automatically, or by a specific reminder to the user.

Undo capability should be provided so that the user can "back up" when necessary. Those systems that are most useable allow several undo steps so that users can reverse more than one action.

Readability and Navigation

- Interfaces should be functional, inviting to use, and pleasing to the eye. It is better to keep to minimalistic designs and to have white space that provide separation between various elements on the user interface.
- It is important to make the text readable by type, size, and color. Some systems allow the user to adjust these properties. Sometimes designers will try to be too fancy and end up with text that is very difficult to read – see example below.

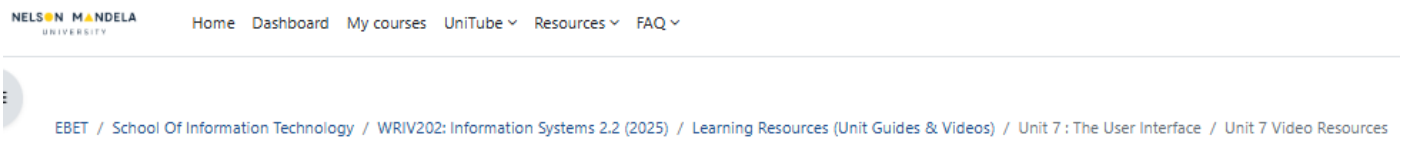


Use of Color and patterns are also important aspects to consider

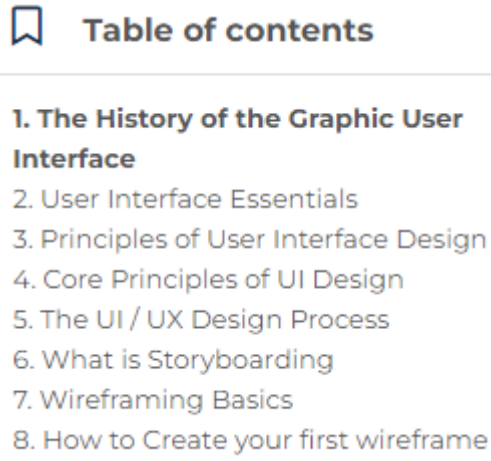
e.g., don't use red on blue

Navigation through the entire application should be obvious and easily traversed. And reversal of navigation, in addition to reversal of actions, should also be provided. One common technique use on websites

specifically, is called **breadcrumbs**, which shows the hierarchy of screens (see example below from the WRIV202 Funda page) that were traversed to arrive at the location for the videos for this unit.



Watch the videos (1 – 4) under Unit 7 Video Resources.
The videos will provide a good introduction and overview of *User Interface Design.*



The Transition from Analysis to User-Interface Design - Considerations for Desktop Applications

- To start the user interface design process; analyst must consider the interaction that the user will have with the system. Often analysts will refer to documentation to assist them with the initial design – this includes the use case narratives as well as the activity diagrams, to name a few. Analysts will typically design use case by se case.

- An analyst will in consultation with the user, create an initial design, wireframe or storyboard that will visually describe the interaction / functionally of a specific use case. A powerful technique in the design of the user interface is to use storyboards. Storyboards are nothing more than a set of layouts that mimic the sequence of steps for a particular use case or user process.

Another important aspect of the user interface that needs to be considered is the Menu Layout and options that will be provided to the users. Menus are a way of grouping large numbers of related use cases or dialogs within a user interface. In all but the smallest systems, menus are needed to present the user with a tractable number of choices per screen, to group related functions together so users can more easily locate them.

A designer must decide which use cases and user interfaces to include in which menus, which menus are required, and how many menu levels are required. Menus usually contain five or so choices in order to avoid overloading the user. Multiple levels of menus helps to keep the number of individual menu items at a workable number.

Use cases with common actors and event decomposition or that implement CRUD actions for a specific domain class, are good candidates to be grouped into a single menu or related group of menus.

Watch the videos (5 - 8) under Unit 7 Video Resources.

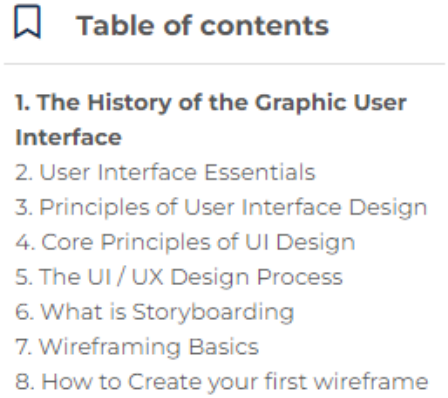


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Look at the document "***UI Design Guidelines***" for some tips and guidelines for appropriate use and examples of HIOs.